

AP PHYSICS 2: ALGEBRA-BASED • THE UNIVERSAL STUDY GUIDE

Targeting a 3, 4, or 5 — every concept tagged for your tier

Tier badges: [3] foundational · [4] adds for 4+ · [5] adds for 5

Targeting a 4 → study [3] AND [4] items. Targeting a 5 → study all three tiers.

PAGE 1 | MATH OF 3/4/5 • 8 UNITS • THERMO + E&M + OPTICS

About this exam. AP Physics 2 picks up where Physics 1 stops — fluids, thermodynamics, electricity, magnetism, electromagnetic waves, geometric and physical optics, and modern physics. Algebra-based throughout. Calculator + reference table. **Conceptual reasoning + paragraph arguments still dominate**, but with much more diagram interpretation (circuit diagrams, ray diagrams, field lines, P-V diagrams).

THE MATH OF YOUR TARGET

Target	MCQ (40q, 50%)	FRQ (4q, 50%)	Approx %	Tier
3 (PASS)	~17 / 40 (~43%)	~16/45 raw	~40-45%	[3]
4	~22 / 40 (~55%)	~24/45 raw	~52-58%	[3] + [4]
5	~28 / 40 (~70%)	~32/45 raw	~68-75%	All

Time: 80 min for 40 MCQ (~2:00 each), 100 min for 4 FRQs (~25 min each). Same FRQ types as Physics 1: math routine, translation, lab/experiment, paragraph argument.

THE 8 UNITS — 2024+ redesign

#	Unit	Weight	Focus
1	Fluids	10–12%	Density, pressure, buoyancy, continuity, Bernoulli
2	Thermodynamics	12–18%	Kinetic theory, P-V diagrams, 1st & 2nd laws, entropy
3	Electric Force, Field, Potential	15–18%	Coulomb's law, E fields, V, equipotentials
4	Electric Circuits	15–18%	Ohm's law, series + parallel, Kirchhoff's laws, RC circuits
5	Magnetism & Electromagnetic Induction	12–15%	B fields, force on charges/wires, Faraday, Lenz
6	Geometric Optics	12–15%	Mirrors, lenses, ray diagrams, refraction
7	Waves, Sound, Physical Optics	10–12%	Standing waves, Doppler, interference, diffraction
8	Modern Physics	5–8%	Photoelectric effect, atomic models, nuclear, mass-energy

KEY INSIGHT [5] Units 3-5 (electricity + magnetism) = ~45% of the exam. Master Coulomb, fields, circuits, induction. Most FRQs sit here.

PAGE 2 | UNIT 1: FLUIDS — DENSITY · PRESSURE · BUOYANCY · FLOW

Density + Pressure [3]/[4]

$$\rho = m/V \quad (\text{density, kg/m}^3)$$

$$P = F/A \quad (\text{pressure, Pa} = \text{N/m}^2)$$

$$\text{Pressure at depth } h \text{ in fluid: } P = P_0 + \rho gh$$

$$\text{Atmospheric pressure: } P_0 \approx 1.01 \times 10^5 \text{ Pa}$$

$$\text{Density of water: } 1000 \text{ kg/m}^3$$

Buoyancy — Archimedes' principle [3]/[4]

$$F_{\text{buoyant}} = \rho_{\text{fluid}} \cdot V_{\text{displaced}} \cdot g$$

$$\text{Floating object: } F_b = mg \rightarrow \rho_{\text{fluid}} \cdot V_{\text{disp}} \cdot g = \rho_{\text{obj}} \cdot V_{\text{obj}} \cdot g$$

$$\text{Fraction submerged} = \rho_{\text{obj}} / \rho_{\text{fluid}}$$

$$\text{Sinking: } F_b < mg. \text{ Apparent weight} = mg - F_b.$$

$$\text{Object floats: } \rho_{\text{obj}} < \rho_{\text{fluid}}$$

Continuity + Bernoulli [3]/[4]/[5]

CONTINUITY (incompressible flow, no leaks):

$$A_1 v_1 = A_2 v_2 \quad (\text{volume flow rate constant})$$

Smaller cross-section $\rightarrow$ faster flow

BERNOULLI (along a streamline, ideal fluid):

$$P_1 + \frac{1}{2}\rho v_1^2 + \rho gh_1 = P_2 + \frac{1}{2}\rho v_2^2 + \rho gh_2$$

Faster flow $\rightarrow$ LOWER pressure (counter-intuitive!).

Why airplanes lift, why curveballs curve, why shower curtains pull in.

KEY INSIGHT [4] Bernoulli intuition. Where flow speeds up (narrow pipe, top of wing), pressure DROPS. The pressure difference does work on the fluid to accelerate it. Energy is conserved per unit volume.

Common fluid traps [3]/[4]

- [3] Pressure depends only on DEPTH, not shape. Same depth in a thin tube vs. wide pool = same pressure.
- [3] Iceberg ~89% submerged: $\rho_{\text{ice}}/\rho_{\text{seawater}} \approx 0.89$.
- [4] Reading P-vs-depth graphs: linear with slope ρg + intercept P_0 .

PAGE 3 | UNIT 2: THERMODYNAMICS — KINETIC THEORY · P-V · LAWS

Kinetic Theory of Gases [3]/[4]

IDEAL GAS LAW: $PV = nRT = NkT$
 n = moles, $R = 8.31 \text{ J/(mol}\cdot\text{K)}$
 N = molecules, $k = 1.38 \times 10^{-23} \text{ J/K}$ (Boltzmann)

AVERAGE KINETIC ENERGY (per molecule): $KE_{\text{avg}} = (3/2) kT$

RMS SPEED: $v_{\text{rms}} = \sqrt{3kT/m} = \sqrt{3RT/M}$

KEY POINT: Temperature IS proportional to average KE of molecules.
 Same $T \rightarrow$ same average KE regardless of molecule type.

First Law of Thermodynamics [3]/[4]

$\Delta U = Q - W$
 ΔU = change in internal energy (T-related for ideal gas)
 Q = heat added TO the gas (positive in)
 W = work done BY the gas (positive when gas expands)

For ideal gas: $\Delta U = (3/2) nR \Delta T$ (monatomic)
 $\Delta U = (5/2) nR \Delta T$ (diatomic, near room T)

P-V diagram processes [3]/[4]/[5]

Process	What's constant	Path on PV	ΔU , Q, W
Isothermal	T (so $\Delta U = 0$)	Hyperbola ($PV = \text{const}$)	$Q = W$
Isobaric	P	Horizontal	$Q = \Delta U + W$; $W = P\Delta V$
Isochoric	V (so $W = 0$)	Vertical	$Q = \Delta U$
Adiabatic	$Q = 0$	Steeper than isotherm	$W = -\Delta U$

WORK on PV DIAGRAM: $W_{\text{by gas}} = \text{AREA UNDER curve}$
 Gas EXPANDS (V increases, $\rightarrow$ right): $W > 0$ (gas does positive work)
 Gas COMPRESSES (V decreases, $\rightarrow$ left): $W < 0$ (work done ON gas)

CYCLES: net W per cycle = AREA enclosed by loop
 Clockwise loop = positive net W (heat engine, $Q_{\text{in}} > Q_{\text{out}}$)
 Counterclockwise = negative (refrigerator)

Second Law + Entropy [4]/[5]

- [4] **2nd Law:** heat flows from hot to cold spontaneously. Entropy of isolated system never decreases.
- [4] **Heat engine efficiency:** $e = W_{\text{net}} / Q_{\text{in}} = 1 - Q_{\text{out}} / Q_{\text{in}}$.
- [5] **Carnot efficiency** (max possible): $e_{\text{max}} = 1 - T_{\text{cold}} / T_{\text{hot}}$ (Kelvin!).
- [5] **Entropy direction:** gas expanding into vacuum, ice melting, hot/cold mixing — all increase total S .

PAGE 4 | UNIT 3: ELECTRIC FORCE · FIELD · POTENTIAL

Coulomb's Law + Electric Field [3]/[4]

COULOMB: $F = k|q_1q_2| / r^2$ $k = 8.99 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2$

ELECTRIC FIELD (force per charge): $E = F/q = kQ/r^2$ (from point charge)

Direction: AWAY from + charge, TOWARD - charge

Force on test charge q : $F = qE$

SUPERPOSITION: E from multiple charges adds as VECTORS.

Field lines: start on +, end on -, density $\propto$ field strength,
never cross.

Electric Potential (Voltage) [3]/[4]

ELECTRIC POTENTIAL: $V = kQ/r$ (scalar! No direction)

POTENTIAL DIFFERENCE (the only thing that matters physically):

$$\Delta V = V_b - V_a$$

WORK to move charge q from A to B: $W = q\Delta V = q(V_b - V_a)$

POTENTIAL ENERGY: $U = qV = kq_1q_2/r$ (scalar; can be $\pm$)

RELATIONSHIP between E and V :

For uniform field: $E = \Delta V/d$ (V drops along field direction)

More general: $E = -dV/dr$

Equipotential surfaces [4]/[5]

- [4] Lines of constant V . Always PERPENDICULAR to E field lines.
- [4] No work done moving along an equipotential ($W = q\Delta V = 0$ since $\Delta V = 0$).
- [5] Conductors in electrostatic equilibrium are equipotential surfaces. E inside = 0; E just outside is perpendicular to surface.

Capacitors [3]/[4]/[5]

$Q = CV$ (charge stored = capacitance $\times$ voltage)

Parallel plate: $C = \epsilon_0 A/d$ (vacuum); with dielectric κ : $C = \kappa \epsilon_0 A/d$

STORED ENERGY: $U = \frac{1}{2} CV^2 = \frac{1}{2} Q^2/C = \frac{1}{2} QV$

Capacitors COMBINING:

Series: $1/C_{eq} = 1/C_1 + 1/C_2$ (smaller than smallest)

Parallel: $C_{eq} = C_1 + C_2$ (just add)

(Note: capacitor combination rules are OPPOSITE of resistors)

PAGE 5 | UNIT 4: CIRCUITS — OHM · KIRCHHOFF · RC

Ohm's Law + Power [3]/[4]

$$V = IR \quad I = \text{current (A)}, \quad R = \text{resistance } (\Omega)$$

$$\text{POWER: } P = IV = I^2R = V^2/R$$

$$\text{ENERGY (over time): } E = Pt$$

$$\text{Resistor properties: } R = \rho L/A \quad (\rho = \text{resistivity, } L = \text{length, } A = \text{cross-section})$$

Series vs Parallel Resistors [3]/[4]

	Series	Parallel
R_eq	$R_1 + R_2 + \dots$	$1/R_{eq} = 1/R_1 + 1/R_2 + \dots$
Current	Same through each	Splits; sums to total
Voltage	Splits; sums to total	Same across each
Add a resistor	R_eq increases	R_eq decreases

WATCH OUT [3] Capacitors and resistors are **OPPOSITE**. Resistors series add; capacitors series reciprocal-add. Don't mix up.

Kirchhoff's Laws [3]/[4]

$$\text{KCL (junction): } \sum I_{in} = \sum I_{out} \quad (\text{charge conservation})$$

$$\text{KVL (loop): } \sum \Delta V = 0 \text{ around any closed loop } (\text{energy conservation})$$

SIGN CONVENTIONS in KVL:

- + going through battery from - to +
- going through battery from + to -
- going through resistor in direction of assumed current (drop)
- + going through resistor against assumed current

RC Circuits [4]/[5]

CHARGING (capacitor through resistor from battery):

$$Q(t) = Q_{max}(1 - e^{(-t/\tau)}) \quad I(t) = (V/R) e^{(-t/\tau)}$$

$$\tau = RC \quad (\text{time constant})$$

DISCHARGING (capacitor through resistor):

$$Q(t) = Q_0 e^{(-t/\tau)} \quad I(t) = (Q_0/RC) e^{(-t/\tau)}$$

At $t = 0$ (start of charging): cap acts like WIRE ($V_{cap} = 0$)

At $t = \infty$ (long time): cap acts like OPEN (no current through)

KEY INSIGHT [4] Initial vs. steady state for capacitors. Just after switch closes: $V_{cap} = 0$, max current. After long time: $V_{cap} = V_{battery}$, $I = 0$ in capacitor branch. Use these to simplify circuits.

PAGE 6 | UNIT 5: MAGNETISM · INDUCTION

Magnetic Force on Charges + Wires [3]/[4]

FORCE on moving charge: $F = qvB \sin \theta$
 Direction: RIGHT-HAND RULE
 Point fingers along v , curl toward $B \rightarrow$ thumb is F (for + charge)
 For - charge: reverse direction

FORCE on current-carrying wire: $F = BIL \sin \theta$
 L = length, I = current, B = field magnitude

CHARGED PARTICLE in uniform B (perpendicular): CIRCULAR motion

Radius: $r = mv/(qB)$

Period: $T = 2\pi m/(qB)$ (independent of v !)

WATCH OUT [3] Magnetic force is always **PERPENDICULAR** to **velocity**. Magnetic force does **NO WORK** on a charge — it only changes direction, not speed.

Magnetic Field Sources [3]/[4]

STRAIGHT WIRE: $B = \mu_0 I / (2\pi r)$ $\mu_0 = 4\pi \times 10^{-7} \text{ T}\cdot\text{m/A}$
 Field circles wire (right-hand grip rule).

SOLENOID (long, n turns/length): $B = \mu_0 nI$
 Uniform inside, ~zero outside (like bar magnet).

Current loop = magnetic dipole. Two parallel currents:
 Same direction = ATTRACT. Opposite = REPEL.

Faraday's Law + Lenz's Law [4]/[5]

MAGNETIC FLUX: $\Phi_B = B \cdot A \cdot \cos \theta$
 θ = angle between B and area vector (normal to surface)

FARADAY'S LAW: $\mathcal{E} = -d\Phi_B/dt$
 Changing flux INDUCES an EMF (and current if there's a loop).

LENZ'S LAW (the negative sign):
 Induced current OPPOSES the change in flux.
 If flux INCREASING into page $\rightarrow$ induced current makes B OUT of page
 $\rightarrow$ current goes COUNTERCLOCKWISE (right-hand grip).

How to change Φ : change B , change A (size), or change θ (rotate).

KEY INSIGHT [5] Lenz's Law conceptual core. A change in flux is what drives induction. The induced current produces its OWN B that fights the change. This is conservation of energy at work — you have to do work against the opposing force to keep the change going.

PAGE 7 | UNITS 6-7: GEOMETRIC + PHYSICAL OPTICS · WAVES · SOUND

Reflection + Refraction [3]/[4]

REFLECTION: $\theta_{\text{incident}} = \theta_{\text{reflected}}$ (both measured from NORMAL)

SNELL'S LAW (refraction): $n_1 \sin \theta_1 = n_2 \sin \theta_2$

$n = c/v_{\text{medium}}$ (index of refraction)

Vacuum: $n = 1$. Water: $n \approx 1.33$. Glass: $n \approx 1.5$.

TOTAL INTERNAL REFLECTION (going dense $\rightarrow$ less dense):

Critical angle: $\sin \theta_c = n_2/n_1$ ($n_1 > n_2$)

Beyond θ_c , all light reflects. Used in fiber optics.

Mirrors + Lenses (Thin Lens Equation) [3]/[4]/[5]

THIN LENS / MIRROR: $1/d_o + 1/d_i = 1/f$

MAGNIFICATION: $M = -d_i/d_o = h_i/h_o$

SIGN CONVENTIONS:

$f > 0$: converging lens or concave mirror

$f < 0$: diverging lens or convex mirror

$d_i > 0$: REAL image (same side as light goes – opposite side of lens / front of mirror)

$d_i < 0$: VIRTUAL image (same side as object on lens / behind mirror)

$M < 0$: inverted image | $M > 0$: upright image

$|M| > 1$: enlarged | $|M| < 1$: reduced

Ray Diagrams — converging lens [4]

- **Ray 1 [4]**: parallel to axis $\rightarrow$ through focal point on far side.
- **Ray 2 [4]**: through center of lens $\rightarrow$ undeflected.
- **Ray 3 [4]**: through near focal point $\rightarrow$ exits parallel to axis.
- **Image** = where rays converge (real) OR appear to diverge from (virtual).

Wave Properties [3]/[4]

$v = f\lambda$ (speed = frequency $\times$ wavelength)

$T = 1/f$ (period = 1/frequency)

STANDING WAVES on string (fixed both ends): $f_n = nv/(2L)$ $n = 1, 2, 3\ldots$

STANDING WAVES in pipe open both ends: $f_n = nv/(2L)$ $n = 1, 2, 3\ldots$

STANDING WAVES in pipe closed one end: $f_n = nv/(4L)$ $n = 1, 3, 5\ldots$ (odd only)

DOPPLER (qualitative for AP P2):

Source approaching observer $\rightarrow$ higher f , shorter λ

Source receding $\rightarrow$ lower f , longer λ

Interference + Diffraction (Physical Optics) [4]/[5]

DOUBLE-SLIT (Young's experiment):

Constructive (bright): $d \sin \theta = m\lambda$ $m = 0, \pm 1, \pm 2\ldots$

Destructive (dark): $d \sin \theta = (m + \frac{1}{2})\lambda$

On screen: bright fringe spacing $y = m\lambda L/d$

THIN FILMS (e.g., oil on water, soap bubble):

Constructive/destructive depends on:

(a) film thickness, (b) wavelength in film ($\lambda_{\text{film}} = \lambda/n$),

(c) phase shifts at boundaries ($180^\circ = \pi$ phase shift on dense $\rightarrow$ less dense reflection)

SINGLE-SLIT diffraction: destructive: $a \sin \theta = m\lambda$ ($m = \pm 1, \pm 2\ldots$)

PAGE 8 | UNIT 8: MODERN · FRQ STRATEGY · COMMON TRAPS

Photoelectric Effect [4]/[5]

Photon energy: $E = hf = hc/\lambda$ $h = 6.626 \times 10^{-34} \text{ J}\cdot\text{s}$

Einstein's equation: $KE_{\text{max}} = hf - \phi$

ϕ = work function (energy to free electron)

Below threshold frequency $f_0 = \phi/h$, NO electrons emitted regardless of intensity

Intensity $\uparrow \rightarrow$ MORE electrons (NOT higher KE).

Frequency $\uparrow \rightarrow$ HIGHER KE per electron.

Stopping voltage: $eV_{\text{stop}} = KE_{\text{max}}$

KEY INSIGHT [5] Why photoelectric matters. Classical wave theory predicted intensity should determine KE. Experiment showed FREQUENCY does. $\rightarrow$ Light is QUANTIZED (photons). Won Einstein the Nobel Prize.

Atomic Energy Levels [4]/[5]

Photon emitted/absorbed when electron transitions between levels:

$E_{\text{photon}} = \Delta E = E_{\text{initial}} - E_{\text{final}}$

$f = \Delta E/h$ $\lambda = hc/\Delta E$

EMISSION: electron drops to lower level $\rightarrow$ photon out

ABSORPTION: photon in $\rightarrow$ electron jumps up

Electron in atom can ONLY have specific (quantized) energies.

Continuous spectrum (hot solid) vs. line spectrum (atoms).

Nuclear Physics [4]

Mass-energy equivalence: $E = mc^2$

(Mass DEFECT $\times c^2$ = binding energy of nucleus.)

ALPHA decay (α): loses ${}^4\text{He}$ nucleus $\rightarrow A-4, Z-2$

BETA-MINUS (β^-): $n \rightarrow p + e^- + \bar{\nu}$ $\rightarrow A$ same, $Z+1$

BETA-PLUS (β^+): $p \rightarrow n + e^+ + \nu$ $\rightarrow A$ same, $Z-1$

GAMMA (γ): photon emission, no change in A or Z

CONSERVATION: charge (Z), nucleon number (A), energy, momentum

FRQ strategy for Physics 2 [3]/[4]/[5]

- [3] Always draw a diagram first — circuit, ray, P-V, field lines, FBD. The diagram organizes everything.
- [3] Identify what's CONSERVED — charge in circuits, energy in collisions/cycles, flux when no change.
- [4] Connect macro to micro — pressure from molecular collisions, current from drift velocity, induced EMF from changing flux.
- [5] Paragraph argument: state law by name (Coulomb's, Lenz's, Faraday's, Bernoulli's), apply to scenario, derive conclusion, interpret.

Common Physics 2 traps [3]/[4]/[5]

- [3] Capacitors series/parallel are OPPOSITE of resistors.
- [3] V is scalar, E is vector — adding voltages = simple sum; adding fields = vector sum.
- [4] Magnetic force does no work — only changes direction.
- [4] Lenz's law direction: induced current opposes the CHANGE, not the field itself.
- [4] Image distance sign: real $\rightarrow$ positive (light passes through location); virtual $\rightarrow$ negative (light only appears to come from there).
- [5] Photoelectric: intensity controls NUMBER of electrons; frequency controls KE. Trapping question.

PAGE 9 | MASTER EQUATION SHEET — by topic

Fluids

$$P = P_0 + \rho gh \quad F_b = \rho Vg \quad A_1 v_1 = A_2 v_2$$

$$P + \frac{1}{2} \rho v^2 + \rho gh = \text{const (Bernoulli)}$$

Thermodynamics

$$PV = nRT = NkT \quad KE_{\text{avg}} = (3/2)kT \quad v_{\text{rms}} = \sqrt{3kT/m}$$

$$\Delta U = Q - W \quad W = P\Delta V \text{ (isobaric)} \quad W = \text{area on PV}$$

$$e = W_{\text{net}} / Q_{\text{in}} \quad e_{\text{carnot}} = 1 - T_c/T_h$$

Electrostatics

$$F = kq_1 q_2 / r^2 \quad E = F/q = kQ/r^2 \quad V = kQ/r$$

$$U = qV = kq_1 q_2 / r \quad E = \Delta V/d \text{ (uniform)} \quad W = q\Delta V$$

$$Q = CV \quad C = \epsilon_0 A/d \quad U_{\text{cap}} = \frac{1}{2} CV^2$$

Circuits

$$V = IR \quad P = IV = I^2 R = V^2/R$$

$$\text{Series: } R_{\text{eq}} = \Sigma R \quad \text{Parallel: } 1/R_{\text{eq}} = \Sigma (1/R_i)$$

$$\text{KCL: } \Sigma I_{\text{in}} = \Sigma I_{\text{out}} \quad \text{KVL: } \Sigma \Delta V = 0 \text{ around loop}$$

$$\tau = RC \quad Q(t) = Q_{\text{max}}(1 - e^{-(t/\tau)}) \text{ (charging)}$$

Magnetism + Induction

$$F = qvB \sin \theta \quad F = BIL \sin \theta \quad r = mv/(qB)$$

$$B_{\text{wire}} = \mu_0 I/(2\pi r) \quad B_{\text{solenoid}} = \mu_0 nI$$

$$\Phi_B = BA \cos \theta \quad \text{EMF} = -d\Phi/dt \text{ (Faraday + Lenz)}$$

Optics

$$n_1 \sin \theta_1 = n_2 \sin \theta_2 \quad \sin \theta_c = n_2/n_1$$

$$1/d_o + 1/d_i = 1/f \quad M = -d_i/d_o = h_i/h_o$$

$$d \sin \theta = m\lambda \text{ (double slit, constructive)}$$

$$f_n = nv/(2L) \text{ (string, pipe both open)}$$

$$f_n = nv/(4L) \text{ (pipe one closed, } n \text{ odd)}$$

Modern

$$E_{\text{photon}} = hf = hc/\lambda \quad KE_{\text{max}} = hf - \phi$$

$$\Delta E = E_i - E_f = hf \text{ (atomic transitions)}$$

$$E = mc^2 \text{ (mass-energy)}$$

PAGE 10 | SCORE LADDER · TIER CHECKLISTS · THE ONE THING

THE SCORE LADDER

For a 3 (Pass)	For a 4	For a 5
MCQ: 17-21 / 40	MCQ: 22-26 / 40	MCQ: 28-32 / 40
Fluids (buoyancy), 1st law thermo, Coulomb, Ohm's law solid.	Add: P-V cycles, Kirchhoff loops, RC time constant, Snell's law, Faraday/Lenz.	Mastery: thin films, photoelectric quantitative, capacitor energy in RC, full Bernoulli problems.
Read circuit diagrams + ray diagrams.	Construct ray diagrams from scratch.	Multi-loop Kirchhoff with battery + capacitor + resistor combos.
1-2 paragraph arguments with simple law citation.	Paragraph argument naming Faraday/Lenz/Bernoulli explicitly.	Conceptual paragraphs connecting micro (molecules, electrons, photons) to macro (pressure, current, intensity).

TIER-SPECIFIC TEST-DAY CHECKLISTS

For a 3 — focus only on these:

- ✓ MCQ: 17+ correct.
- ✓ Buoyancy: $F_b = \rho V g$; *floating fraction* = $\rho_{\text{obj}}/\rho_{\text{fluid}}$.
- ✓ PV = nRT for ideal gas problems.
- ✓ Coulomb's law and $E = kQ/r^2$ for point charges.
- ✓ $V = IR$; series adds, parallel reciprocal-adds (resistors).
- ✓ Snell's law and $1/d_o + 1/d_i = 1/f$.
- ✓ Right-hand rule for magnetic force on positive charge.

For a 4 — add these on top:

- ✓ MCQ: 22+ correct.
- ✓ P-V diagrams: identify isothermal/isobaric/isochoric/adiabatic; $W = \text{area}$.
- ✓ Capacitors: series/parallel rules OPPOSITE of resistors; $U = \frac{1}{2} CV^2$.
- ✓ Kirchhoff's loop method for multi-loop circuits.
- ✓ Faraday's law: $\text{EMF} = -d\Phi/dt$; Lenz: induced current opposes change.
- ✓ Ray diagrams for converging/diverging lenses + mirrors.
- ✓ Standing waves: which harmonics for which boundary conditions.

For a 5 — add these on top:

- ✓ MCQ: 28+ correct.
- ✓ Bernoulli with full energy-per-volume reasoning.
- ✓ Carnot efficiency and 2nd-law/entropy reasoning.
- ✓ RC charging/discharging with time constant + initial/final states.
- ✓ Magnetic force does NO work; only changes direction.
- ✓ Photoelectric: frequency vs. intensity quantitative.
- ✓ Connect micro (molecules, electrons, photons) to macro (T, I, intensity).

THE ONE THING THAT MATTERS MOST [3][4][5]: DRAW THE DIAGRAM AND IDENTIFY THE CONSERVATION LAW. Every Physics 2 problem reduces to one of: charge conservation (Kirchhoff), energy conservation (1st law / Bernoulli / collisions), flux change (Faraday/Lenz), or momentum/charge in a field. **Draw the relevant diagram → state which conservation law applies → write equation in this scenario → solve.** Skipping the diagram is the #1 reason students miss FRQs they could otherwise solve.

If you do nothing else: master P-V diagrams, Kirchhoff's laws, and the Faraday/Lenz combo. Those three carry the bulk of FRQ points.

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